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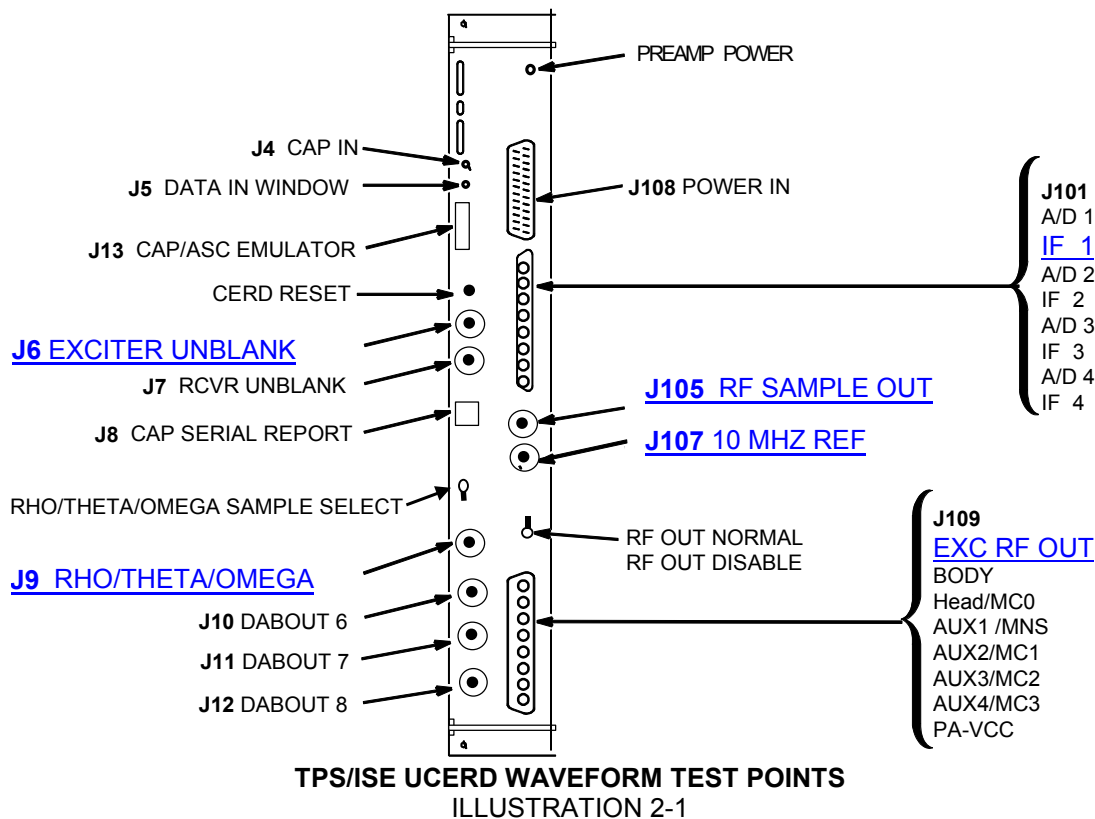
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1- REQUIRED TOOLS

- TPS RF Connector/Adapter and Cable Test Kit, 46-301927G1
- RF Test Cables Kit 46-251710G4
- 100 mHz oscilloscope, Tektronix® 468 or equivalent, 46-183029P61 or P64

2- DESCRIPTION

1. The testpoints to scope for specific waveforms are shown in Illustration 2-1 for both 1.0T and 1.5T systems. Click on the hyperlinks to see scope waveforms from systems that are NOT running Release 9.0, CNV4, HFO2, or later releases of software. Waveforms from systems that are running Release 9.0, CNV4, HFO2, or equivalent releases of software will not have the 180 degree RF pulse. See the comments at the beginning of **SECTION 3- EXCITER SCAN PREPARATION** for more information.



2. The IPG Board Scope Trigger, MR2A30A7J8, is used whenever an external trigger signal is required. Use the BNC to SMB test cable 46-301549P5 (part of TPS RF Connector/Adapter and Cable Test Kit) whenever this is required.
3. RF signals (versus logic signals) require a 50Ω input termination. If the scope does not have a 50Ω input selection, attach a 50Ω in-line termination (part of RF Test Cables Kit) at the scope channel input and set it to 1MΩ (high impedance) input. Check the specific channel setups on each waveform Illustration.
4. Each scope waveform contains the proper channel input selections, time base and triggering information.

3- EXCITER SCAN PREPARATION

Starting with Release 9.0, CNV4, and HFO2 software, only the 90 degree RF waveform will be visible on the oscilloscope. The 180 degree RF waveform will not be visible. This was done to simplify the RF power calibration process by pre-setting the CVs "ia_rf1" and "ia_rf2" to 32766 (maximum amplitude) and 0 (zero; minimum amplitude), respectively. The software level can be determined by opening a c-shell and typing **whatRev** <Enter>. The software level is displayed on the last line of the resulting output. The RF waveforms displayed in **SECTION 5- WAVEFORM DIAGRAMS FOR SYSTEMS NOT RUNNING RELEASE 9.0, CNV4, HFO2, OR LATER VERSIONS OF SOFTWARE** are for systems that have not received the latest software.

The RF waveforms displayed in **SECTION 6- WAVEFORM DIAGRAMS FOR SYSTEMS RUNNING RELEASE 9.0, CNV4, HFO2, OR LATER VERSIONS OF SOFTWARE** are for systems that have received the latest software.

1. Disconnect the cable to the CERD Module, RF EXCITER OUT, MR2A30A13J109.
2. At keypad on front magnet enclosure, press LANDMARK and ADV TO SCAN.
3. At Operator Work space, select the scan icon in the desktop control panel, if you have not already done so.
4. Click on **[New Pt]**, and enter
Id: **geservice**
Name: **rf waveform**
Weight (Lb): **111**
Set Patient Protocols to **Service**.
5. In the Protocol field, type **o.6.1** (o=Other, 1=series) to load protocol.
6. Click on **[Save Series]**.
7. **[Research Operations]**, **[Setup Params]** with the right mouse button, enter settings per the following
R1 = **13**
R2 = **14**
TG = **200**
Number of Frames: **2 <Enter>**

Note

The **<Enter>** key must be pressed after the numeric entries in order for the entries to be saved.

WINDOW 1

Frame:1 **<Enter>**

Frame:0 **<Enter>**

WINDOW 2

Frame:1 **<Enter>**

Frame:0 **<Enter>**

[Done]

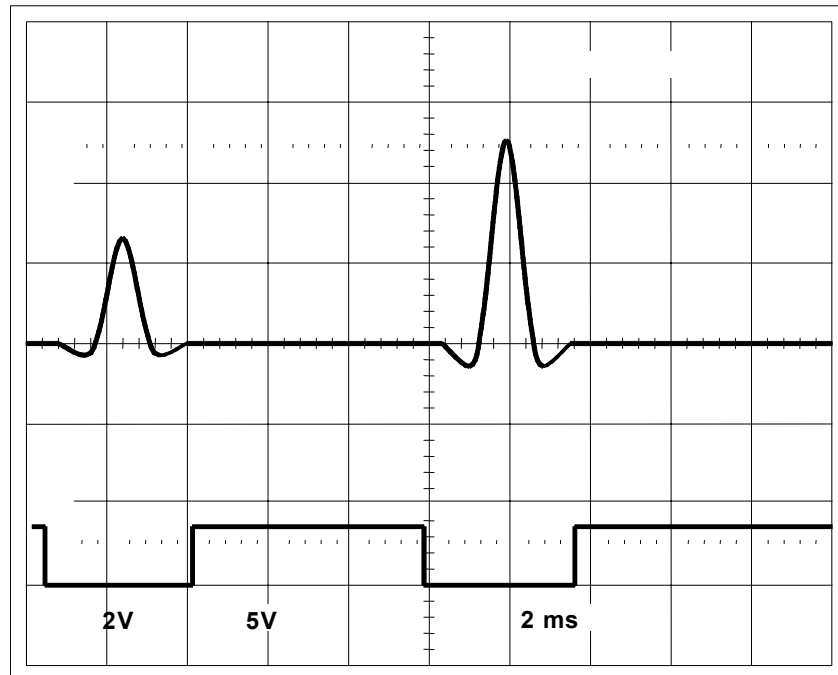
3- EXCITER SCAN PREPARATION (CONTINUED)

8. Set calmode= **5** <Enter> (SINC Pulse), select **[Accept]**.
9. Select **[Manual Prescan]**, **[Scan TR]**
10. Refer to the appropriate section for RF waveforms.
 - a. If the system is NOT operating with Release 9.0, CNV4, HFO2 or later software: Refer to **SECTION 5- WAVEFORM DIAGRAMS FOR SYSTEMS NOT RUNNING RELEASE 9.0, CNV4, HFO2, OR LATER VERSIONS OF SOFTWARE.**
 - b. If the system is operating with Release 9.0, CNV4, HFO2 or later software: Refer to **SECTION 6- WAVEFORM DIAGRAMS FOR SYSTEMS RUNNING RELEASE 9.0, CNV4, HFO2, OR LATER VERSIONS OF SOFTWARE.**

4- RECEIVER SCAN PREPARATION

1. Set pismode= **3** <Enter> (Receiver Loopback), select **[Accept]**.
2. Select **[Manual Prescan]**.
3. Refer to the appropriate section for RF waveforms.
 - a. If the system is NOT operating with Release 9.0, CNV4, HFO2 or later software: Refer to **SECTION 5- WAVEFORM DIAGRAMS FOR SYSTEMS NOT RUNNING RELEASE 9.0, CNV4, HFO2, OR LATER VERSIONS OF SOFTWARE.**
 - b. If the system is operating with Release 9.0, CNV4, HFO2 or later software: Refer to **SECTION 6- WAVEFORM DIAGRAMS FOR SYSTEMS RUNNING RELEASE 9.0, CNV4, HFO2, OR LATER VERSIONS OF SOFTWARE.**

5- WAVEFORM DIAGRAMS FOR SYSTEMS NOT RUNNING RELEASE 9.0, CNV4, HFO2, OR LATER VERSIONS OF SOFTWARE



RHO OUT SAMPLE (J9)

Scope Channel 1:
1M Ω input
2V/div

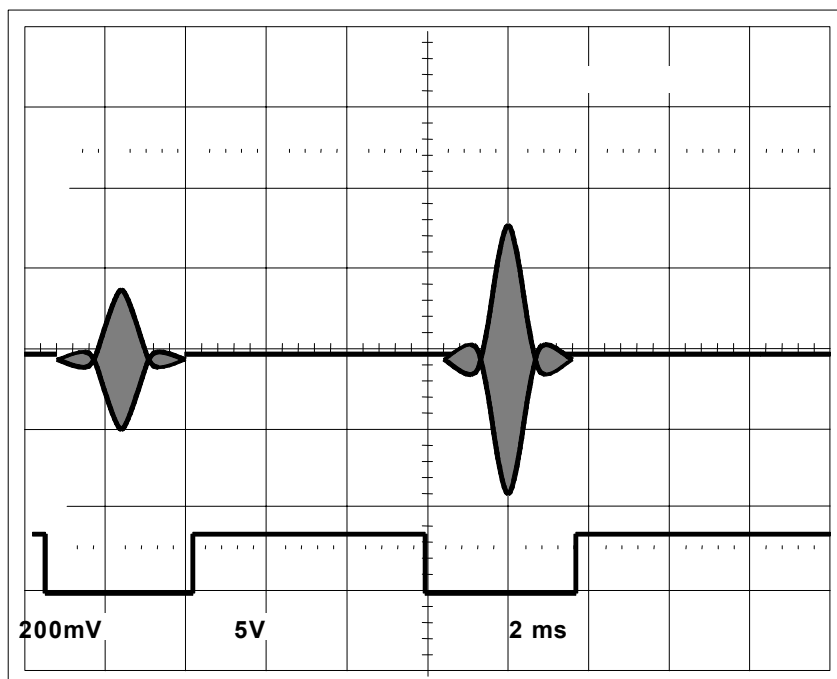
EX UNBLK* (J6)

Scope Channel 2:
1M Ω input
5V/div

Trigger: **←NOTE**
External

**RHO OUT SAMPLE (J9) WAVEFORM
ILLUSTRATION 5-1**

[Click Here to return
to Illustration 2-1](#)



RF SAMPLE OUT (J105)

Scope Channel 1:
50 Ω input **← NOTE**
100mV/div

EX UNBLK* (J6)

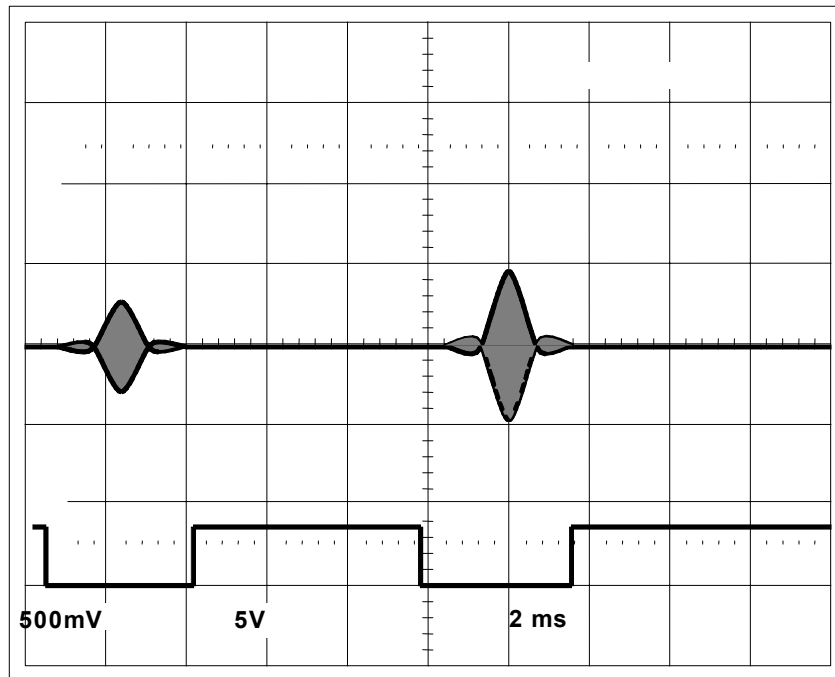
Scope Channel 2:
1M Ω input
5V/div

Trigger: **← NOTE**
External

**RF SAMPLE OUT WAVEFORM (J105)
ILLUSTRATION 5-2**

[Click Here to return
to Illustration 2-1](#)

**5- WAVEFORM DIAGRAMS FOR SYSTEMS NOT RUNNING RELEASE 9.0, CNV4, HFO2,
OR LATER VERSIONS OF SOFTWARE (CONTINUED)**



EXCITER RF OUT (J109)

Scope Channel 1:
50 Ω input **← NOTE**
500mV/div

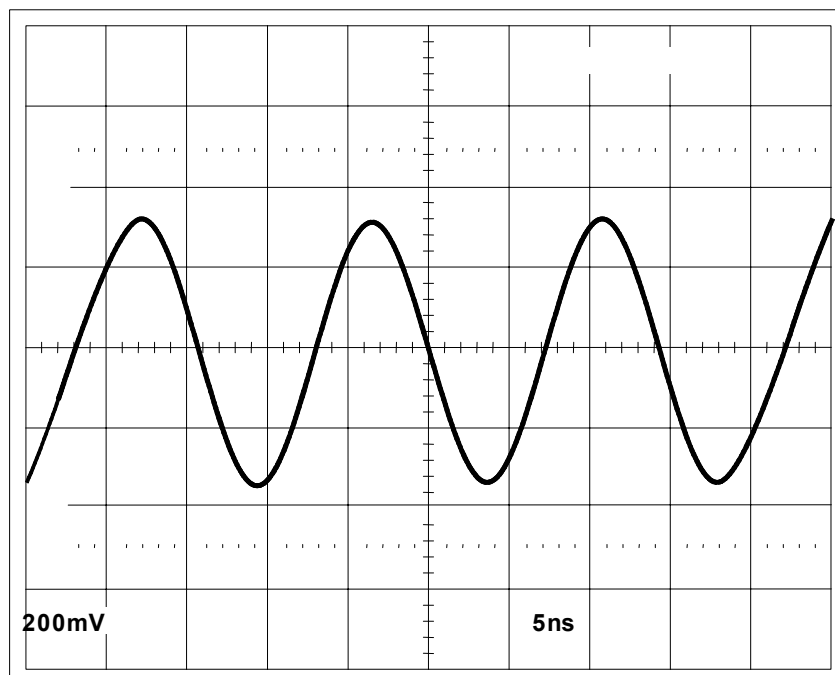
EX UNBLK* (J6)

Scope Channel 2:
1M Ω input
5V/div

Trigger:
External

EXCITER RF OUT WAVEFORM (J109)
ILLUSTRATION 5-3

[Click Here to return
to Illustration 2-1](#)



10 MHZ REF LO1 (J107)

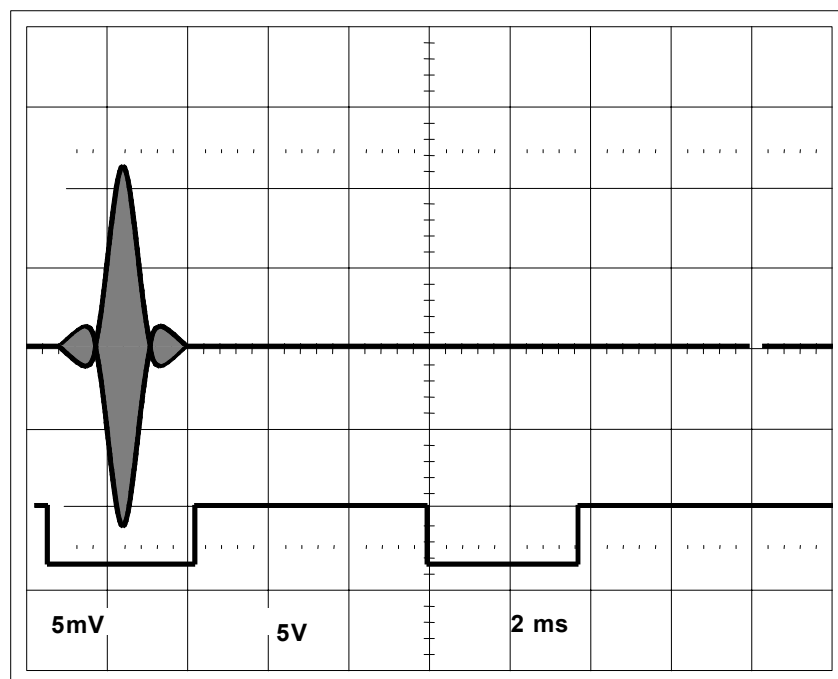
Scope Channel 1:
50 Ω input
200mV/div

Trigger:
Channel 1

10 MHZ REF LO1 (J107)
ILLUSTRATION 5-4

[Click Here to return
to Illustration 2-1](#)

**5- WAVEFORM DIAGRAMS FOR SYSTEMS NOT RUNNING RELEASE 9.0, CNV4, HFO2,
OR LATER VERSIONS OF SOFTWARE (CONTINUED)**



I.F. SAMPLE (IF1)

Scope Channel 1:
50 Ω input
5mV/div

EX UNBLK* (J6)

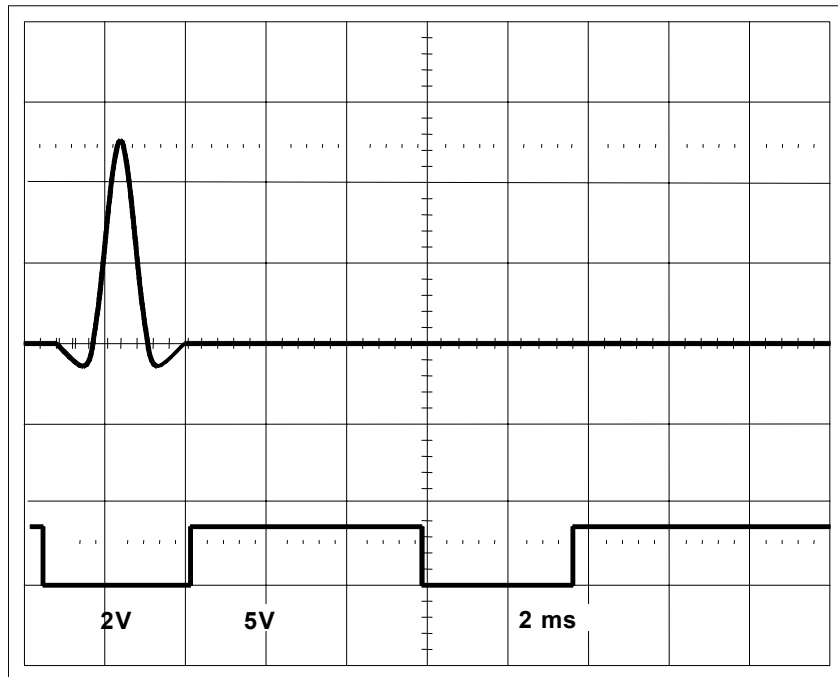
Scope Channel 2:
1M Ω input
5V/div

Trigger:
External

**I.F. SAMPLE (IF1)
ILLUSTRATION 5-5**

[Click Here to return
to Illustration 2-1](#)

6- WAVEFORM DIAGRAMS FOR SYSTEMS RUNNING RELEASE 9.0, CNV4, HFO2 OR EQUIVALENT SOFTWARE



RHO OUT SAMPLE (J9)

Scope Channel 1:
1M Ω input
2V/div

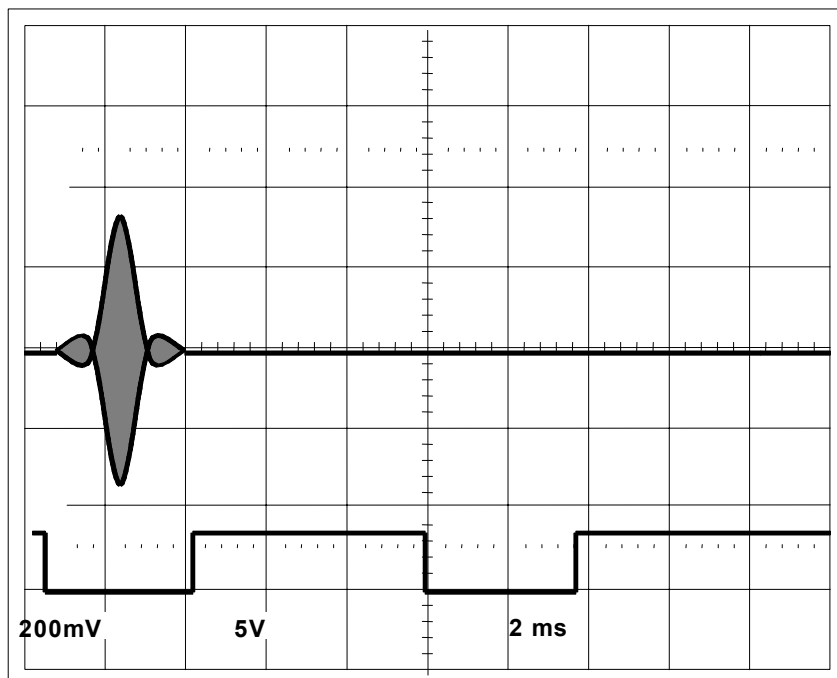
EX UNBLK* (J6)

Scope Channel 2:
1M Ω input
5V/div

Trigger: **←NOTE**
External

**RHO OUT SAMPLE (J9) WAVEFORM
ILLUSTRATION 6-1**

[Click Here to return
to Illustration 2-1](#)



RF SAMPLE OUT (J105)

Scope Channel 1:
50 Ω input **← NOTE**
100mV/div

EX UNBLK* (J6)

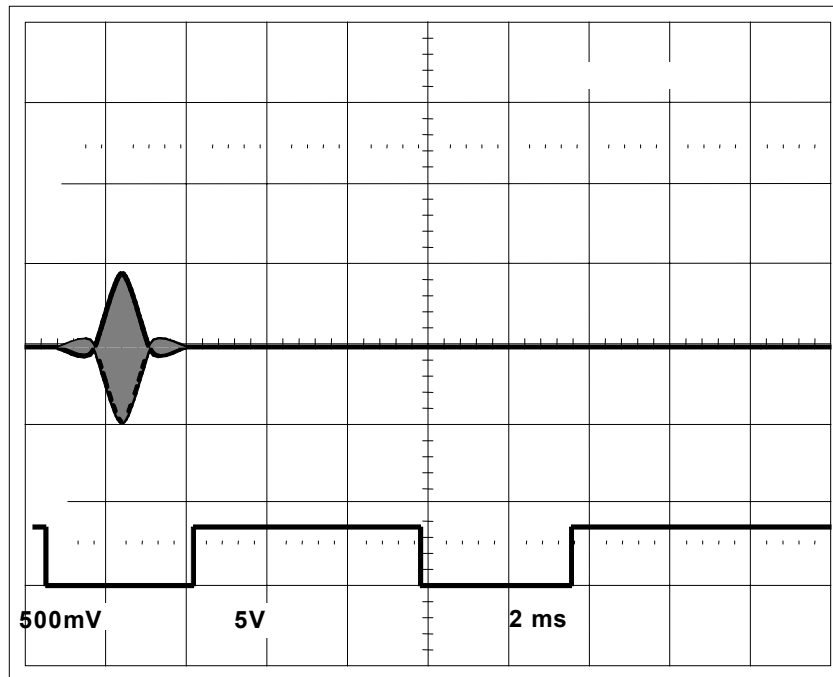
Scope Channel 2:
1M Ω input
5V/div

Trigger: **← NOTE**
External

**RF SAMPLE OUT WAVEFORM (J105)
ILLUSTRATION 6-2**

[Click Here to return
to Illustration 2-1](#)

6- WAVEFORM DIAGRAMS FOR SYSTEMS RUNNING RELEASE 9.0, CNV4, HFO2 OR EQUIVALENT SOFTWARE (CONTINUED)



EXCITER RF OUT (J109)

Scope Channel 1:
50 Ω input **← NOTE**
500mV/div

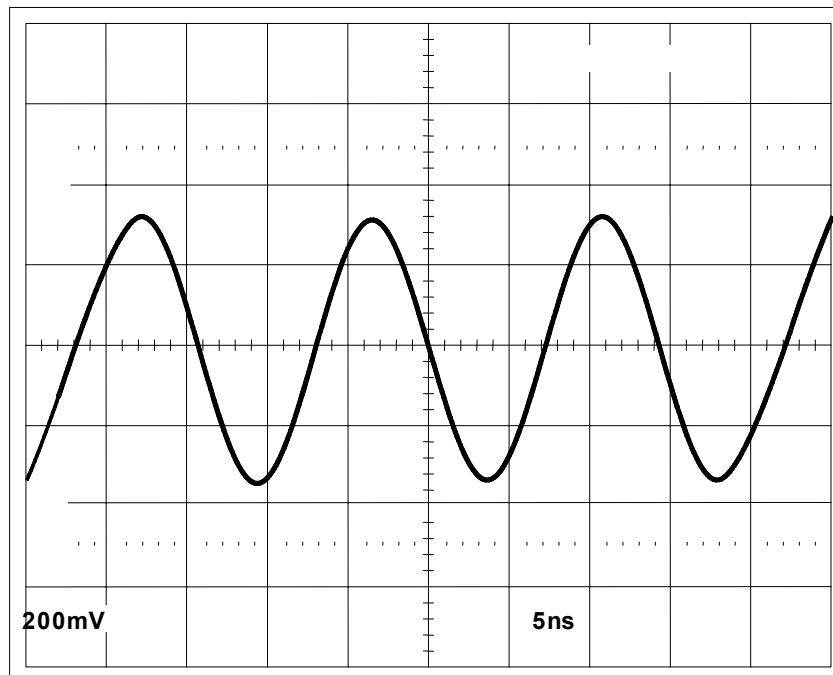
EX UNBLK* (J6)

Scope Channel 2:
1M Ω input
5V/div

Trigger:
External

EXCITER RF OUT WAVEFORM (J109)
ILLUSTRATION 6-3

[Click Here to return to Illustration 2-1](#)



10 MHz REF LO1 (J107)

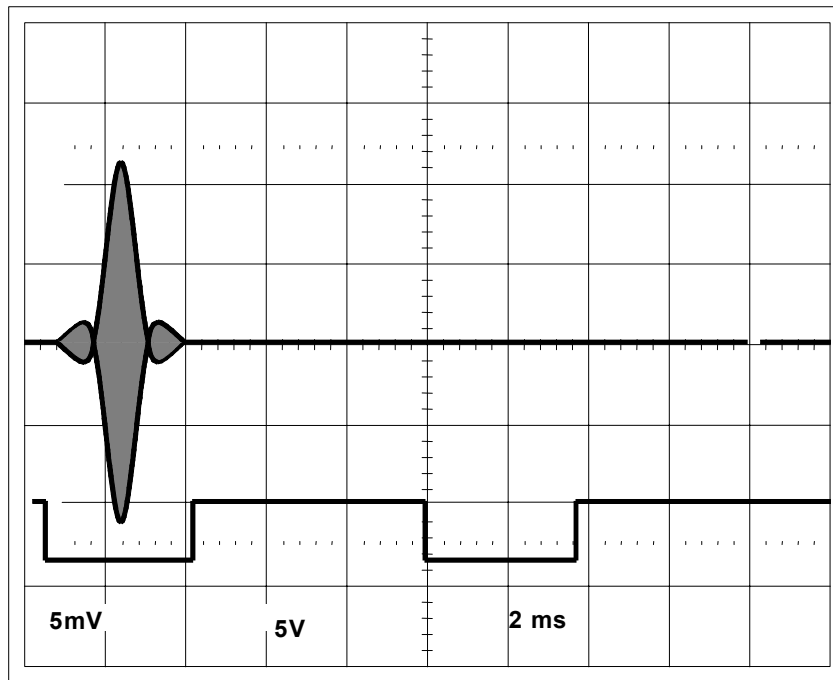
Scope Channel 1:
50 Ω input
200mV/div

Trigger:
Channel 1

10 MHz REF LO1 (J107)
ILLUSTRATION 6-4

[Click Here to return to Illustration 2-1](#)

6- WAVEFORM DIAGRAMS FOR SYSTEMS RUNNING RELEASE 9.0, CNV4, HFO2 OR EQUIVALENT SOFTWARE (CONTINUED)



I.F. SAMPLE (IF1)

Scope Channel 1:
50 Ω input
5mV/div

EX UNBLK* (J6)

Scope Channel 2:
1M Ω input
5V/div

Trigger:
External

I.F. SAMPLE (IF1)
ILLUSTRATION 6-5

[Click Here to return
to Illustration 2-1](#)

7- SYSTEM RESTORATION

1. Reconnect the cable to the CERD module, RF EXCITER OUT, MR2A30A13J109.
2. Ensure that all system cables are installed.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 2, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Nov 12, 1998	M. Keber	Removed obsolete 8.1 information; misc. style guide cleanup.
2	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.
3	Feb. 22, 2000	R. Kaufman	Added missing waveforms from toolbook conversion.
4	Oct 23, 2001	D. Thome'	Added Illustrations and added information advising of absence of 180 degree RF pulse for systems with 9.0, CNV4, HFO2, or later software.